

ELT3207 Meas. & Digital Control Basics

Lecture 8: Analysis of digital control systems

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Where are we

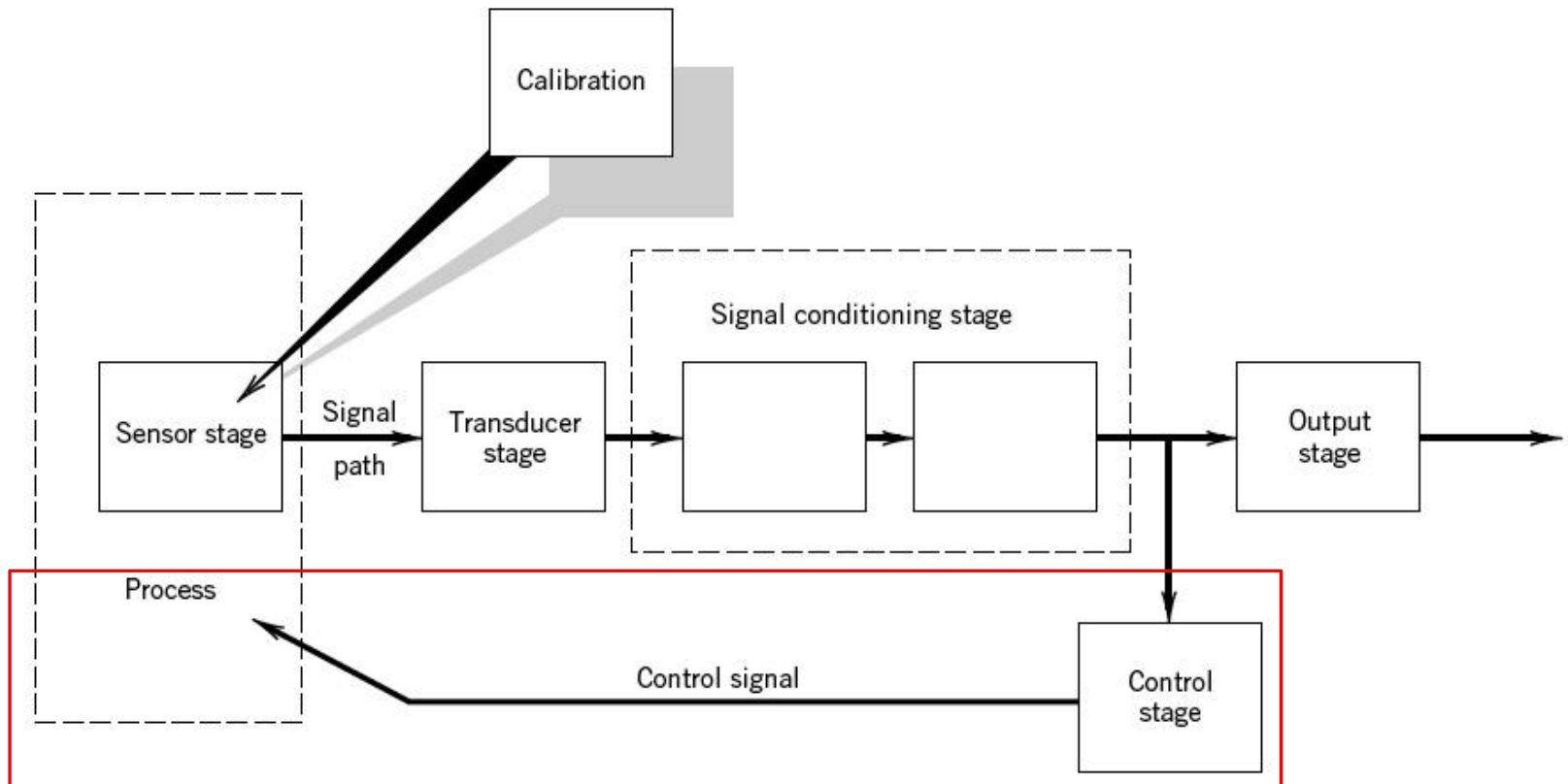


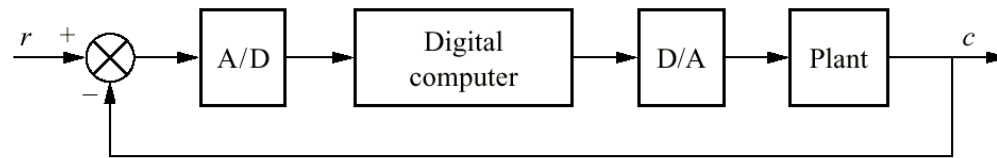
Figure 1.1 Components of a general measurement system.

□ Today's lecture: Analysis of digital control systems

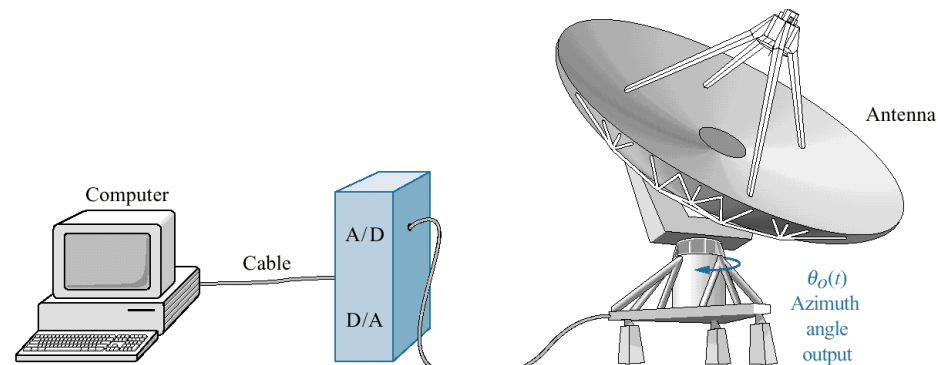
Digital Control Systems

What

- This course focuses on *dynamic* control systems



(b)



(b)

Advantages and disadvantages of digital controllers

Advantages

- ✓ Improved sensitivity.
- ✓ Use digital components → wide application, cost.
- ✓ Control algorithms easily modified → flexibility, speed.
- ✓ Many systems inherently are digital, e.g. radar tracking systems, a space satellite.

Disadvantages

- Develop complex math algorithms.
- Lose information during conversions due to technical problems.
- Most signals are continuous in nature.

Design procedure

❑ Emulation design procedure

1. Start with continuous system (ELT3051)

- Modelling.
- Analysis and synthesis loop.

2. Add sampled-data system elements.

- Sampler
- Zero-order hold.

3. Chose sample period:

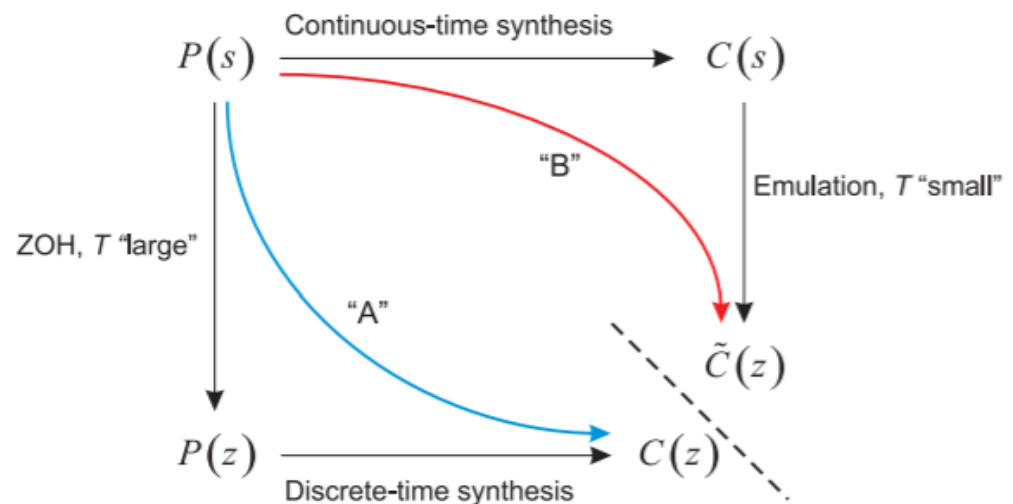
- Usually small but not too small.

4. Digitize control law

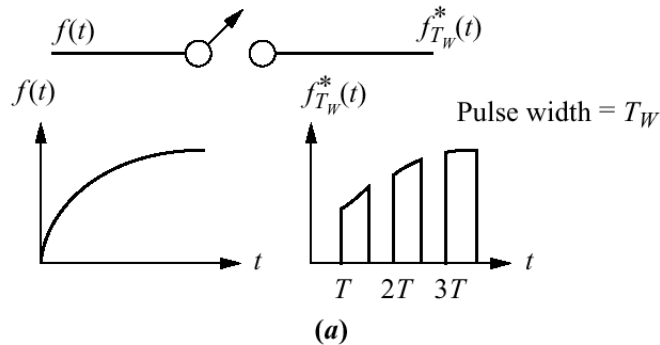
- Differencing methods
- Bilinear transformation

5. Performance verification

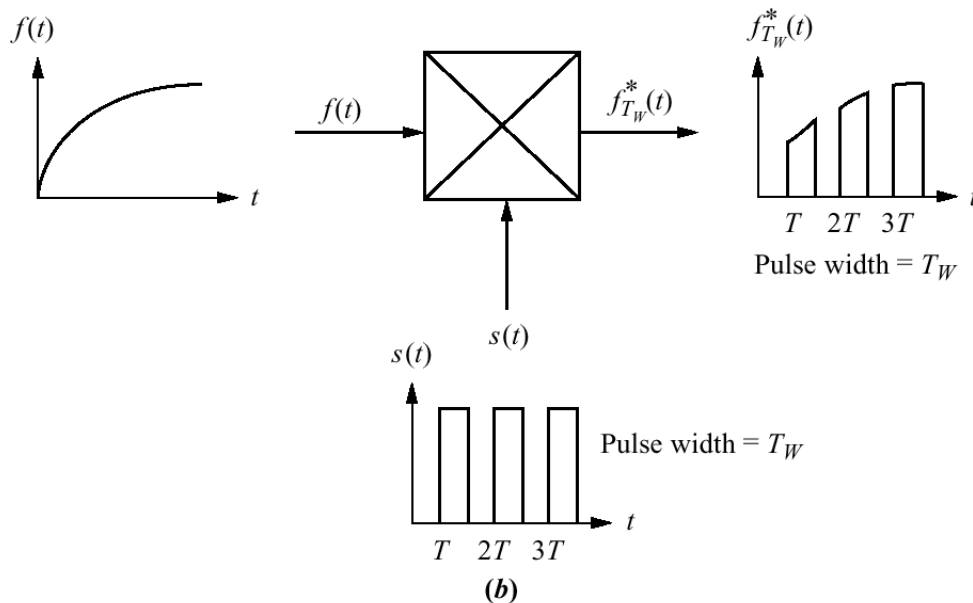
- Simulation



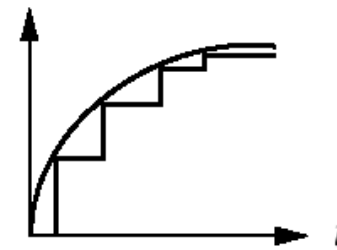
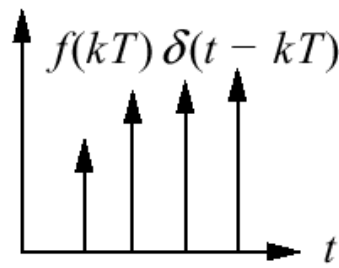
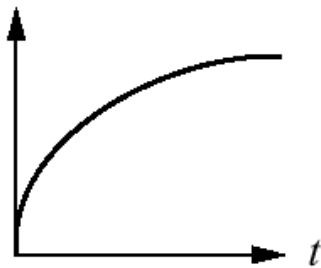
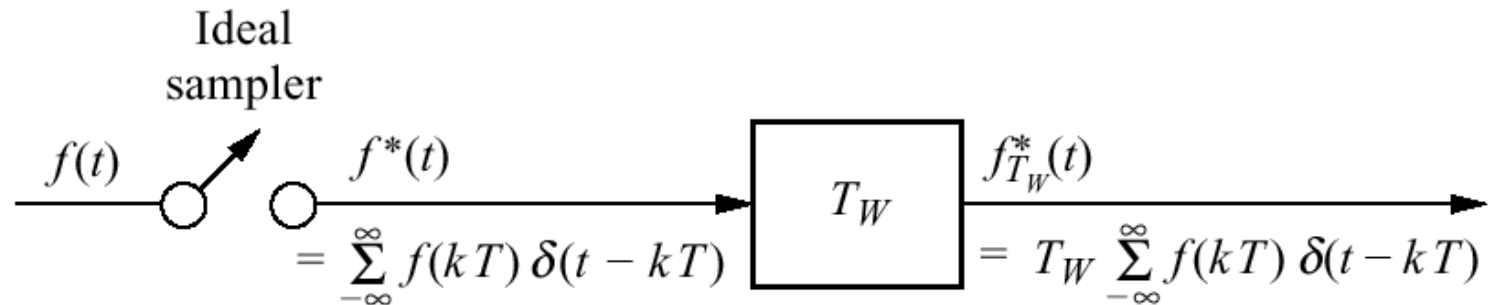
Modeling the Digital Computer: the sampler



Two views of uniform-rate sampling:
a. switch opening and closing;
b. product of time waveform and sampling waveform



Modeling the Digital Computer: zero-order sample-and-hold



❑ Zero-order hold:

- Impulse response is $h(t) = u(t) - u(t - T)$
- Transfer function: $G_h(s) = \frac{1}{s} - \frac{1}{s} e^{-sT} = \frac{1 - e^{-sT}}{s}$

The z-transform (recap)

- ❑ An important tool in the analysis and design of DT systems.
 - Plays a role similar to that served by Laplace transforms in CT systems.
- ❑ Mapping from the s-plane to the z-plane: $z = e^{sT}$
 - $F(z) = Z[f(t)] = \mathcal{L}[f^*(t)]|_{z=e^{sT}} = \sum_{k=0}^{\infty} f(kT)z^{-k}$

	$f(t)$	$F(s)$	$F(z)$	$f(kT)$
1.	$u(t)$	$\frac{1}{s}$	$\frac{z}{z-1}$	$u(kT)$
2.	t	$\frac{1}{s^2}$	$\frac{Tz}{(z-1)^2}$	kT
3.	t^n	$\frac{n!}{s^{n+1}}$	$\lim_{a \rightarrow 0} (-1)^n \frac{d^n}{da^n} \left[\frac{z}{z - e^{-aT}} \right]$	$(kT)^n$
4.	e^{-at}	$\frac{1}{s+a}$	$\frac{z}{z - e^{-aT}}$	e^{-akT}
5.	$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$	$(-1)^n \frac{d^n}{da^n} \left[\frac{z}{z - e^{-aT}} \right]$	$(kT)^n e^{-akT}$
6.	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$	$\frac{z \sin \omega T}{z^2 - 2z \cos \omega T + 1}$	$\sin \omega kT$
7.	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$	$\frac{z(z - \cos \omega T)}{z^2 - 2z \cos \omega T + 1}$	$\cos \omega kT$
8.	$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2 + \omega^2}$	$\frac{ze^{-aT} \sin \omega T}{z^2 - 2ze^{-aT} \cos \omega T + e^{-2aT}}$	$e^{-akT} \sin \omega kT$
9.	$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2 + \omega^2}$	$\frac{z^2 - ze^{-aT} \cos \omega T}{z^2 - 2ze^{-aT} \cos \omega T + e^{-2aT}}$	$e^{-akT} \cos \omega kT$

The Inverse z-Transform (recap)

❑ Inverse z-Transforms via Partial-Fraction Expansion.

➤ Decompose $F(z)$ into $F(z) = \frac{Az}{z-z_1} + \frac{Bz}{z-z_1} + \dots$

➤ Applying $e^{-akT} \xleftrightarrow{Z} \frac{z}{z-e^{aT}}$

❑ Inverse z-Transform via the Power Series Method

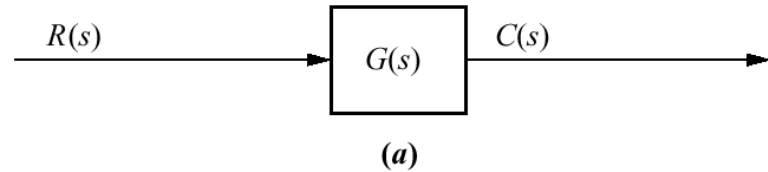
➤ Performing the indicated division to yield a power series for $F(z)$

➤ **Example:** $F(z) = \frac{0.5z}{(z-0.5)(z-0.7)} = \frac{0.5z}{z^2-1.2z+0.35} = 0.5z^{-1} + 0.6z^{-2} + 0.545z^{-3} + \dots$

→ $F^*(s) = 0.5e^{-Ts} + 0.6e^{-2Ts} + 0.545e^{-3Ts} + \dots$

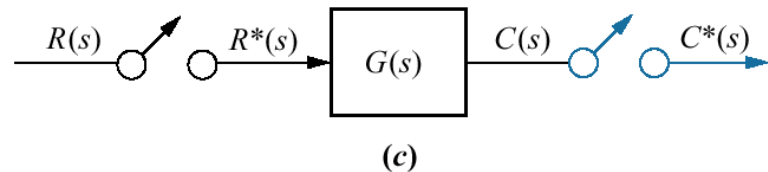
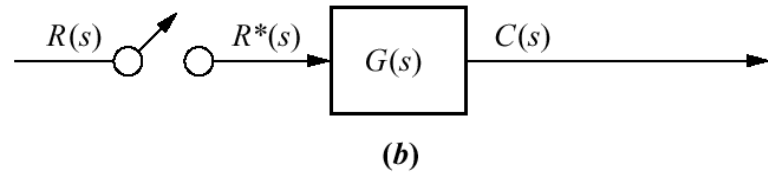
→ $f^*(t) = 0.5\delta(t-T) + 0.6\delta(t-2T) + 0.545\delta(t-3T) + \dots$

Transfer functions of sampled-data systems



Sampled-data systems:

- a. continuous;
- b. sampled input;
- c. sampled input & output (sync'd.)



Note: Phantom sampler is shown in color.

- b) Sampled input: $r^*(t) = \sum_{n=0}^{\infty} r(nT)\delta(t - nT)$; continuous output: $c(t) = \sum_{n=0}^{\infty} r(nT)g(t - nT)$
- c) If we focus on the output at the sampling instants and **not** in between: $c(kT) = \sum_{n=0}^{\infty} r(nT)g(kT - nT) \rightarrow$ representation of sampled system is greatly simplified: $C(z) = \sum_{k=0}^{\infty} c(kT)z^{-k} = G(z)R(z) \rightarrow G(z) = C(z)/R(z)$: **pulse transfer function**.

Example

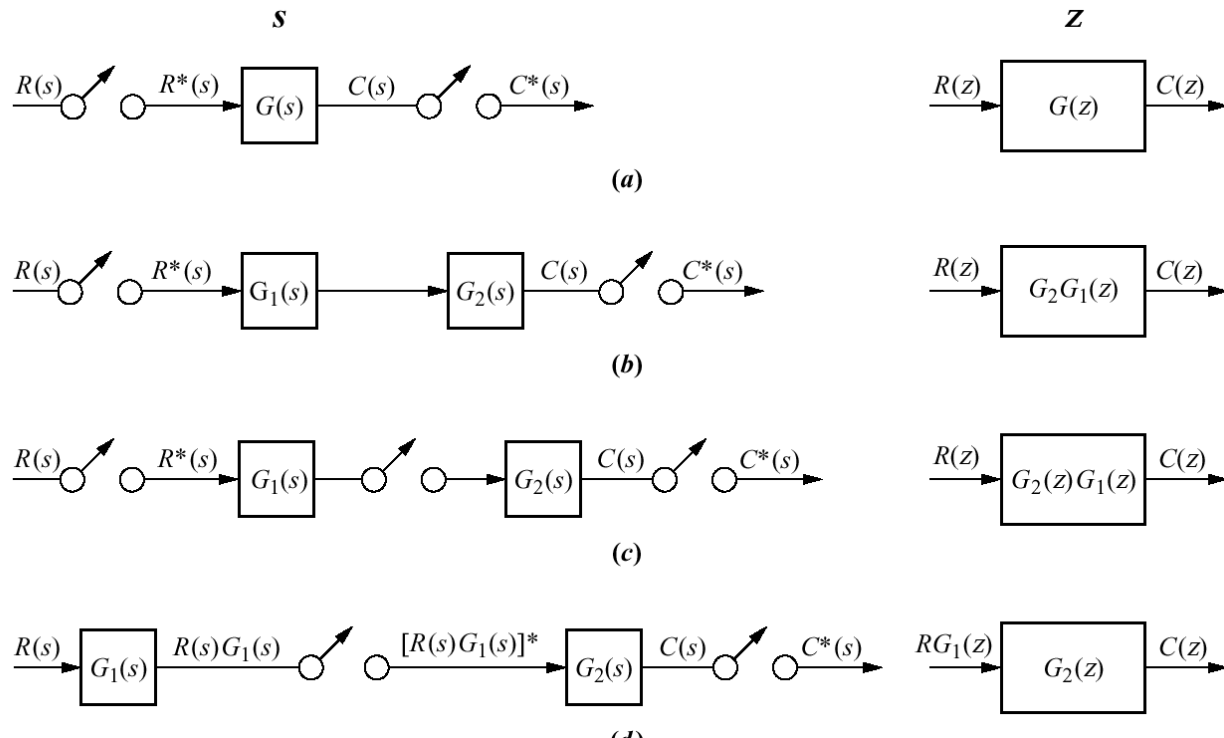
- Given a z.o.h. in cascade with $G_1(s) = \frac{s+2}{s+1}$, or $G(s) = \frac{1-e^{-sT}}{s} \frac{s+2}{s+1}$, find the sampled-data transfer function, $G(z)$, if the sampling time, T , is 0.5 second.

□ **Solution:**

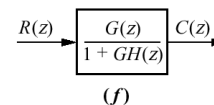
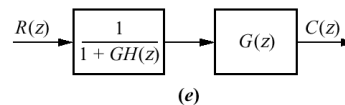
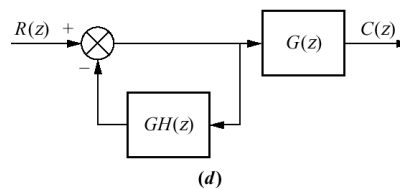
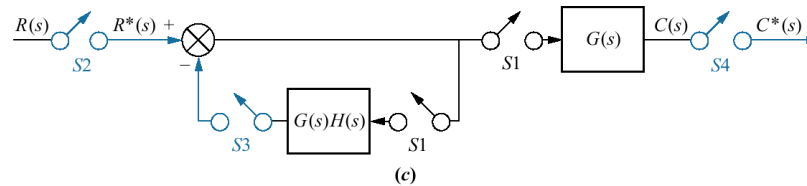
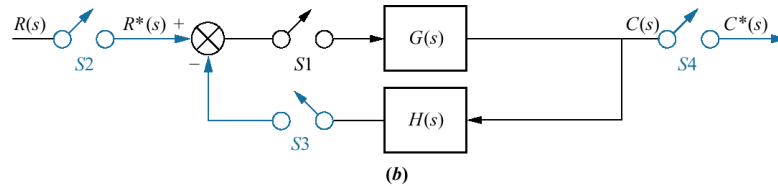
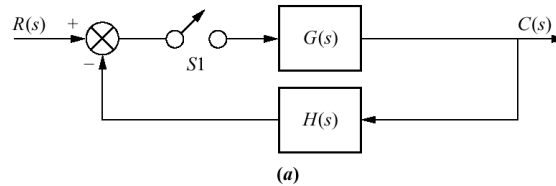
- $G(s) = (1 - e^{-sT}) \frac{G_1(s)}{s} \rightarrow G(z) = (1 - z^{-1}) \mathcal{Z} \left\{ \frac{G_1(s)}{s} \right\}$
- Let $G_2(s) = \frac{G_1(s)}{s} = \frac{s+2}{s(s+1)} = \frac{A}{s} + \frac{B}{s+1} = \frac{2}{s} - \frac{1}{s+1}$
- Taking the inverse Laplace transform: $g_2(t) = 2 - e^{-t} \rightarrow g_2(kT) = 2 - e^{-kt}$
- Using the z-transform table: $G_2(z) = \frac{2z}{z-1} - \frac{z}{z-e^{-T}}$. Substituting $T = 0.5$ yields
$$G_2(z) = \frac{2z}{z-1} - \frac{z}{z-0.607} = \frac{z^2 - 0.213z}{(z-1)(z-0.607)}.$$
- Hence $G(z) = \frac{z-1}{z} G_2(z) = \frac{z-0.213}{z-0.607}$

Block diagram reduction (1)

- Care must be taken to avoid mistakes, e.g. $\mathcal{Z}\{G_1(s)G_2(s)\} \neq G_1(z)G_2(z)$

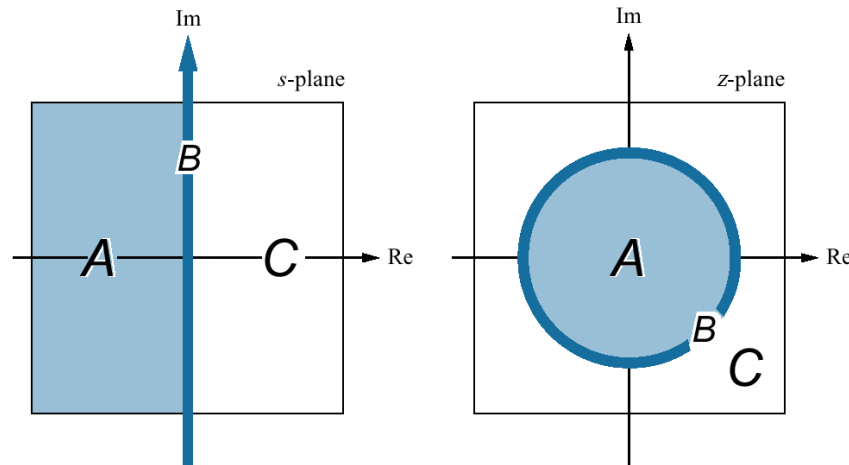


Block diagram reduction (2)



Note: Phantom samplers are shown in color.

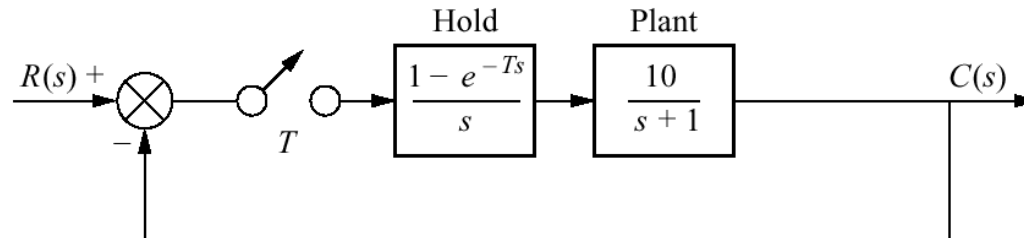
Stability analysis (recap)



- ❑ Let $s = \sigma + j\omega$
 - Since $z = e^{Ts} \rightarrow z = e^{\sigma T} \angle \omega T \rightarrow$ each region of the s-plane can be mapped into a corresponding region on the z-plane
- ❑ Digital System Stability via the z-Plane
 - **stable** if all poles of the closed-loop transfer function are **inside** the unit circle on the z-plane
 - **unstable** if any pole is **outside** the unit circle and/or there are poles of **multiplicity greater than one** on the unit circle
 - **marginally stable** if poles of multiplicity one are on the unit circle and all other poles are inside the unit circle

Example

- ❑ Determine the ranges of sampling interval, T , that will make the system shown below stable and unstable.



❑ Solution:

- The closed-loop transfer function $T(z) = \frac{G(z)}{1+G(z)}$
- Perform partial-fraction expansion on $G(s)$: $G(s) = 10(1 - e^{-Ts}) \left(\frac{1}{s} - \frac{1}{s+1} \right)$
- Taking the z-transform, we obtain $G(z) = 10 \frac{1-e^{-T}}{z-e^{-T}}$
- Substituting $10 \frac{1-e^{-T}}{z-e^{-T}}$ for $G(z)$ in the first equation yields $T(z) = \frac{10(1-e^{-T})}{z-(11e^{-T}-10)}$, hence:
 - ✓ The pole of $T(z)$ will be inside the unit circle, i.e. the system will be stable if $0 < T < 0.2$
 - The system will be unstable if $0.2 < T$

Bilinear transformation

- ❑ Allow us to apply our s-plane analysis and design techniques (e.g. Routh-Hurwitz criterion), then convert the results to a digital system that contains the same properties.
- ❑ Bilinear trans. is of the form $z = \frac{as+b}{cs+d}$ & its inverse is $s = \frac{-dz+b}{cz+a}$
 - Different values of a, b, c , and d have been derived for particular applications and yield various degrees of accuracy
 - The transformations are not exact → only the property for which they are designed can be relied upon, e.g. we **cannot** expect $G(s)$ to have the same **transient response** as $G(z)$ if a, b, c, d are chosen for **stability** transformation.
- ❑ Digital System Stability via the s-Plane: choose $s = \frac{z+1}{z-1}$ & its inverse $z = \frac{s+1}{s-1}$
 - **$j\omega$ -axis** points on the s-plane will be mapped to **unit-circle** points on the z-plane, **right-half-plane** points on the s-plane to points **outside** the unit circle on the z-plane, and **left-half-plane** points on the s-plane to points **inside** the unit circle on the z-plane.

Example

- Given $T(z) = \frac{N(z)}{D(z)}$, where $D(z) = z^3 - z^2 - 0.2z + 0.1$, use the Routh-Hurwitz criterion to find the number of z-plane poles of $T(z)$ inside, outside, and on the unit circle. Is the system stable?

□ **Solution:**

- Substitute $\frac{s+1}{s-1}$ for z into the equation $D(z) = 0$ yields $s^3 - 19s^2 - 45s - 17 = 0$.
- Routh table:

s^3	1	-45
s^2	-19	-17
s^1	-45.89	0
s^0	-17	0

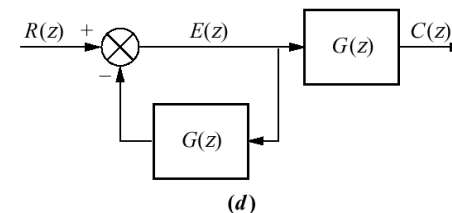
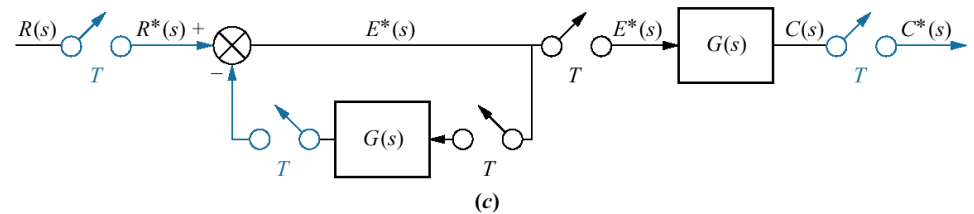
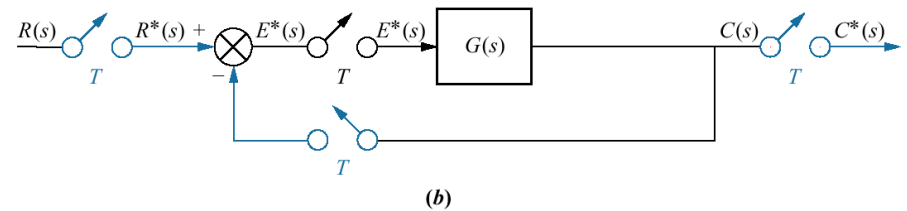
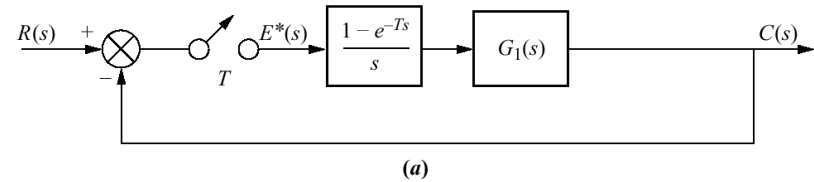
- $T(z)$ has one pole outside the unit circle, no poles on the unit circle, and two poles inside the unit circle → unstable.

Steady-State Error Analysis (1)

- ❑ Generally more difficult than analog case
 - Digital systems: the placement of the sampler changes the open-loop transfer function and thus precludes any general conclusions.

- Assume the typical placement of the sampler after the error and in the position of the cascade controller.

$$\begin{aligned}
 \text{➤ } E(z) &= \frac{R(z)}{1+G(z)} \rightarrow e^*(\infty) = \\
 &\lim_{z \rightarrow 1} (1 - z^{-1}) \frac{R(z)}{1+G(z)}
 \end{aligned}$$



Note: Phantom samplers are shown in color.

Steady-State Error Analysis (2)

□ Unit Step Input:

➤ $R(s) = \frac{1}{s} \rightarrow R(z) = \frac{z}{z-1} \rightarrow e^*(\infty) = \frac{1}{1 + \lim_{z \rightarrow 1} G(z)}.$

➤ Defining the static error constant $K_p = \lim_{z \rightarrow 1} G(z)$, $e^*(\infty) = \frac{1}{1 + K_p}.$

□ Unit Ramp Input.

➤ $R(z) = \frac{Tz}{(z-1)^2} \rightarrow e^*(\infty) = \frac{1}{K_v}$, where $K_v = \frac{1}{T} \lim_{z \rightarrow 1} (z-1)G(z)$

□ Unit Parabolic Input

➤ $R(z) = \frac{T^2 z(z+1)}{(z-1)^3} \rightarrow e^*(\infty) = \frac{1}{K_a}$, where $K_a = \frac{1}{T^2} \lim_{z \rightarrow 1} (z-1)^2 G(z)$

□ Steady-state error equations are similar to the equations developed for analog systems

- Multiple pole placement at $z = 1$ reduces the steady-state error to zero for digital systems of appropriate type.
- Since stability depends upon the sampling interval, be sure to check the stability of the system after a sampling interval is established before making steady-state error calculations.

Summary

- ❑ Advantages of digital control systems:
 - Can control numerous loops at reduced cost.
 - System modifications can be implemented with software changes.
- ❑ The z-transform.
 - Allows us to represent sampled waveforms at the sampling instants
 - Allow handling of sampled systems as easily as continuous systems, including block diagram reduction
- ❑ Stability analysis
 - Can be done in the z-plane: the unit circle becomes the boundary of stability
 - Can also be done in the s-plane via bilinear transformation
- ❑ Steady-state error analysis
 - Depends on the placement of the sampler, but parallels the s-plane techniques.
- ❑ **Next lecture:** Digital controller synthesis.